

CO900G – ENVIRONMENTAL SCIENCE AND SUSTAINABLE ENGINEERING
SECTION B – ALTERNATIVE ASSESSMENT METHOD (AAM)

INDIVIDUAL TECHNICAL REPORT

Environmental Problem:
IMPROPER DISPOSAL OF ELECTRONIC WASTE

Proposed IT Solution:
SMART E-WASTE COLLECTION AND RECYCLING WEBSITE

Main Idea:
A website that checks the condition of old electronic items and recommends whether to reuse, repair, or recycle them, while helping users find a proper collection center.

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Table of Contents

1. Abstract
 2. Introduction
 3. Problem Statement
 4. Objectives of the Proposed Solution
 5. Sustainability Analysis
 6. Proposed Engineering/IT Solution
 7. System Design and Working
 8. Feasibility Analysis
 9. Expected Environmental Benefits
 10. Justification of the Proposed Solution
 11. Limitations and Future Scope
 12. Conclusion
 13. References
- Appendix A – Proposed Website Screens

1. Abstract

Electronic waste (e-waste) is generated when electrical and electronic equipment reaches the end of its useful life or is no longer wanted by its owner. Improper disposal of such equipment can result in loss of valuable materials and can create environmental and health risks because some electronic products contain hazardous substances. In India, e-waste management is regulated under the E-Waste (Management) Rules, 2022, which came into effect on 1 April 2023. The Central Pollution Control Board (CPCB) describes environmentally sound management and Extended Producer Responsibility (EPR) as important elements of the framework [1].

This report proposes a Smart E-Waste Collection and Recycling Website. The proposed system is designed to help ordinary users make a practical decision about an unwanted electronic device before disposal. The user enters the type and condition of an item, such as a mobile phone, laptop, television or charger. The website then provides a simple recommendation: reuse the item if it is still usable, repair/refurbish it if it is damaged but potentially recoverable, or recycle it if it is no longer suitable for use. The system can then help the user locate a suitable collection centre or submit a pickup request.

The main purpose of the solution is to make responsible e-waste management easier and more convenient. Instead of treating every unwanted device as immediate waste, the system introduces a simple decision process based on the item's condition. The solution is aligned mainly with Sustainable Development Goal 12, particularly Targets 12.4 and 12.5, which address environmentally sound waste management and reduction of waste through prevention, reduction, recycling and reuse [2].

The proposed prototype can be implemented using HTML, CSS and JavaScript, as required by the assessment brief. It can later be expanded with a database, map services, user accounts, QR-based tracking and real-time integration with authorized collection and recycling organizations.

2. Introduction

The widespread use of electronic devices has transformed education, communication, employment, entertainment and daily life. Mobile phones, computers, televisions, printers, chargers and other electrical and electronic equipment are now common in homes and institutions. As technology develops quickly, users may replace older products even when the products are still partially functional.

When an electronic product is no longer required, the next decision is important. A working device may have a second user. A damaged laptop may be repairable. A completely non-functional device may need to enter a formal recycling process. If the user simply places the item with ordinary household waste, the opportunity for reuse or material recovery may be lost.

CPCB notes that e-waste contains valuable materials, while some e-waste components can contain heavy metals and other substances that may pose risks during inappropriate handling and recovery [1]. Therefore, e-waste management is not only a question of collecting discarded devices; it also involves making sure that products are directed toward an appropriate end-of-life pathway.

The proposed website addresses this information and accessibility gap. It is intended as a user-facing digital platform rather than a replacement for authorized recycling infrastructure. Its purpose is to guide the user, provide collection information and make the process easier to understand.

The assessment brief requires every student to independently analyze the environmental problem and propose a unique Engineering/IT-based solution, after which the group will evaluate the proposals and

implement one selected solution as a responsive website [3]. The solution presented in this report is therefore intentionally focused on condition-based decision making: reuse, repair/refurbishment or recycling.

3. Problem Statement

3.1 Environmental Problem

The selected environmental problem is the improper disposal of electronic waste. Improper disposal includes discarding electronic items through unsuitable waste channels or failing to use appropriate collection, refurbishment and recycling routes.

3.2 Causes of the Problem

- Limited public awareness about what qualifies as e-waste and how different devices should be handled.
- Difficulty finding convenient and trustworthy collection options.
- Uncertainty about whether an old device can still be reused or repaired.
- Short replacement cycles for some consumer electronics.
- Lack of a simple digital process connecting users with responsible collection pathways.
- Limited visibility of what happens after an electronic item is handed over.

3.3 Environmental and Resource Implications

Improper e-waste handling can result in loss of recoverable materials and inappropriate release or handling of hazardous substances. It also reduces the opportunity to extend the useful life of products through reuse and repair. At the global level, the United Nations reports that e-waste reached 65 million tonnes in 2023, while only 15.5 million tonnes were formally recycled [2]. These figures demonstrate the scale of the management challenge.

3.4 Problem Gap Addressed by the Website

Many awareness approaches stop at the message that electronic waste should be recycled. A user, however, may still ask: Can this device be reused? Can it be repaired? Where should I take it? Can it be collected? The proposed website addresses these practical questions in one interface.

4. Objectives of the Proposed Solution

- To create a simple digital platform for responsible e-waste management.
- To help users classify an unwanted electronic item according to its condition.
- To recommend reuse, repair/refurbishment or recycling as appropriate pathways.
- To make collection-centre information easier to access.
- To provide an optional pickup-request workflow for the prototype.
- To improve awareness of responsible e-waste handling.
- To demonstrate how a web application can support sustainable consumption and waste management.
- To provide a foundation that can later be expanded into a larger e-waste lifecycle management system.

5. Sustainability Analysis

5.1 Sustainable Development Goal 12

The proposed solution is most directly aligned with SDG 12: Responsible Consumption and Production. UN Target 12.4 calls for environmentally sound management of chemicals and all wastes throughout their life cycle, while Target 12.5 calls for substantially reducing waste generation through prevention, reduction, recycling and reuse [2].

SDG Target	Connection with Proposed Website
12.4 – Environmentally sound waste management	The website guides users toward appropriate collection and responsible end-of-life handling.
12.5 – Waste prevention, reduction, recycling and reuse	The decision tool places reuse and repair/refurbishment before recycling when appropriate.
12.8 – Information and awareness	The website provides users with practical information for sustainable choices.

5.2 Circular Economy Perspective

The proposed system follows a simple circular-economy hierarchy. Instead of assuming that an unwanted device has no remaining value, the website first checks whether it can remain in use. If not, it considers repair or refurbishment before recommending recycling. This approach attempts to preserve the value of the product for as long as possible.

The intended pathway is: Buy/Use → Assess Condition → Reuse → Repair/Refurbish → Reuse → Recycle → Material Recovery. The website does not itself perform physical recycling; it provides the digital decision and connection layer.

5.3 ISO 14001 Perspective

ISO 14001 is an international standard for environmental management systems. The current edition is ISO 14001:2026, published in April 2026 [4]. ISO describes the standard as a framework for organizations to manage environmental responsibilities systematically, improve environmental performance and consider a life-cycle perspective.

The proposed website can support, but does not itself provide, an ISO 14001 environmental management system. Its relevance is mainly in the areas of environmental awareness, operational control, monitoring and continual improvement. For example, an institutional version of the portal could track the number and categories of devices collected, reuse/refurbishment outcomes and recycling activity. These records could support environmental objectives and performance monitoring.

5.4 Life-Cycle Perspective

The solution considers the end-of-use stage of an electronic product. It also recognizes that the best outcome may not always be immediate recycling. Extending useful life through reuse or repair can delay the need for a new product and can preserve product value. Where a device is no longer suitable for use, responsible recycling becomes the appropriate pathway.

6. Proposed Engineering/IT Solution

6.1 Solution Name

Smart E-Waste Collection and Recycling Website

6.2 Main Idea

A website that checks the condition of old electronic items and recommends whether to reuse, repair, or recycle them, while helping users find a proper e-waste collection centre.

6.3 Core User Flow

Step	User Action	System Response
1	Select electronic item	Displays relevant item categories and questions.
2	Enter condition	Evaluates simple condition inputs.
3	Receive recommendation	Shows Reuse, Repair/Refurbish or Recycle.
4	Choose collection option	Shows suitable collection information.
5	Submit request	Generates a pickup/collection request.
6	Track status	Displays a simple lifecycle status.

6.4 Functional Modules

E-Waste Checker: Allows users to select an item and enter basic condition information.

Decision Recommendation: Uses predefined JavaScript rules to recommend reuse, repair/refurbishment or recycling.

Collection Centre Finder: Provides a searchable list of collection locations and the categories they accept.

Pickup Request: Collects basic request details and generates a request number for the prototype.

Tracking Screen: Shows stages such as Submitted, Scheduled, Collected and Sent for Processing.

Impact Dashboard: Summarizes the number of items submitted and the pathway selected.

6.5 Uniqueness of the Proposed Solution

The main difference between this solution and a basic e-waste collection portal is the condition-based decision stage. The system does not immediately direct every item to recycling. It first asks whether the product can reasonably continue to be used or repaired. This gives the individual proposal a clear and explainable IT component that can be implemented using JavaScript decision rules.

7. System Design and Working

7.1 Proposed System Architecture

Layer	Main Components
User Interface	Home page, e-waste checker, collection finder, pickup form, tracking and dashboard
Application Logic	JavaScript validation, decision rules and request generation
Data Layer	Prototype data using JavaScript/local storage; future version can use a database
External Services	Optional map/location services and notification services
Output	Recommendation, collection information, request ID and status

7.2 Decision Logic

The first prototype can use a transparent rule-based approach instead of complex artificial intelligence. This makes the system easy to test and explain during a student demonstration.

Condition	Recommendation
Device works normally and is still useful	Reuse
Device works partly or has a repairable fault	Repair / Refurbish
Device is severely damaged or non-functional	Recycle
User is uncertain	Show safe-disposal/collection guidance

7.3 Technology Stack

Technology	Purpose
HTML	Structure of webpages and forms
CSS	Layout, styling and responsive design
JavaScript	Interactive features and decision logic
Local Storage / Database	Store prototype requests and user status
Optional Map API	Show collection locations in a real implementation

The assessment brief specifies HTML, CSS and JavaScript, with frameworks optional, and requires a responsive website with at least four functional webpages/screens and at least two interactive features [3]. The proposed system can therefore be implemented within the stated technical requirements.

7.4 Proposed Pages

- Home – explains the environmental problem and introduces the solution.
- E-Waste Checker – the main interactive condition-assessment feature.
- Collection Finder – helps users identify a suitable collection option.
- Schedule Pickup – allows users to submit a collection request.
- Track E-Waste – displays the request lifecycle.

- Impact Dashboard – shows the user's contribution.

7.5 Interactive Features

Interactive Feature 1: E-Waste Checker. Users enter device category and condition and receive an immediate recommendation.

Interactive Feature 2: Pickup Request. Users enter item and collection details and receive a generated request ID.

Additional interactive features may include collection-centre search, tracking and an impact calculator.

8. Feasibility Analysis

8.1 Technical Feasibility

The solution is technically feasible for a student prototype because its core functions can be implemented with standard web technologies. A basic version does not require specialized hardware. The decision engine can use simple JavaScript conditions, while collection data can initially be stored as a local dataset.

8.2 Economic Feasibility

A prototype can be developed using commonly available development tools. A larger real-world deployment would introduce costs for hosting, databases, maps, notifications, cybersecurity and maintenance. However, the academic prototype can demonstrate the concept without requiring these full-scale services.

8.3 Operational Feasibility

The workflow is designed for ordinary users. The user does not need technical knowledge: they select a device, answer simple condition questions and follow the recommendation. The system can therefore function as a public-facing guidance platform.

8.4 Social Feasibility

Potential users include households, students, educational institutions, offices and small businesses. A simple website may be especially useful in institutions where old computers, chargers, phones and other electronic equipment accumulate.

8.5 Legal and Environmental Considerations

The proposed website should not claim to replace authorized recyclers. In India, the E-Waste (Management) Rules, 2022 establish responsibilities for producers and other stakeholders, including recycling through registered recyclers [1]. A real deployment would therefore need verified collection and recycling partners and appropriate handling of user data.

9. Expected Environmental Benefits

Expected Benefit	How the Solution Can Contribute
More responsible disposal	Provides a clear alternative to ordinary disposal.
Greater reuse	Identifies devices that may still be usable.

Repair/refurbishment	Directs potentially repairable items toward a recovery pathway.
Improved recycling access	Helps users locate suitable collection options.
Awareness	Explains why correct e-waste handling matters.
Data for improvement	Future versions can record trends in device categories and collection demand.

The benefits listed above are expected or potential benefits of the proposed digital system; they should not be presented as measured environmental outcomes until the system is deployed and supported by reliable data.

10. Justification of the Proposed Solution

The proposed solution is justified because the environmental problem has a practical information and accessibility component. Telling users that e-waste should be recycled does not always tell them what to do with a particular device. The Smart E-Waste Collection and Recycling Website converts a general environmental message into a step-by-step decision process.

The most important feature is the condition assessment. If an electronic device is still useful, reuse may be preferable to immediately treating it as waste. If it is damaged but repairable, repair or refurbishment may extend its useful life. If it is no longer suitable for use, the system directs the user toward recycling. This gives the solution a clear logic and differentiates it from a simple collection-directory website.

The proposed solution also fits the assessment requirements. It is an Engineering/IT-based concept, it can be implemented as a responsive website using HTML, CSS and JavaScript, and it can include multiple functional screens and interactive features [3].

The solution is also consistent with India's current e-waste management framework, which includes EPR and recognizes refurbishment and recycling within the management system [1]. The website is therefore best understood as a digital support layer that can help consumers reach appropriate pathways.

11. Limitations and Future Scope

11.1 Limitations

- The student prototype may use sample collection-centre data rather than a live nationwide database.
- Pickup and tracking may be simulated for demonstration purposes.
- The condition assessment is rule-based and depends on the user's answers.
- Environmental impact figures should not be treated as verified measurements unless supported by reliable datasets.
- A production system would require strong privacy, authentication and cybersecurity controls.
- Real-world operation would require coordination with authorized collection, refurbishing and recycling organizations.

11.2 Future Scope

- AI-based image recognition to identify electronic products from photographs.
- QR codes for tracking items through collection and processing stages.
- Real-time integration with authorized recyclers and collection partners.
- Mobile application for easier access and notifications.

- Institutional dashboards for colleges, offices and businesses.
- Reward or certificate systems to encourage responsible disposal.
- Geographic analytics to identify areas with insufficient collection infrastructure.

12. Conclusion

Improper disposal of electronic waste is an environmental and resource-management challenge that requires convenient and understandable solutions for consumers. The proposed Smart E-Waste Collection and Recycling Website addresses this challenge by combining a condition-based decision tool with collection guidance and a simple request and tracking process.

The central idea is simple: an old electronic item should not automatically be treated as waste. The user first checks whether the item can be reused, repaired or refurbished. If it cannot reasonably continue in use, the system guides the user toward appropriate recycling and collection. This creates a clear digital pathway from an unwanted electronic product to a more responsible end-of-life option.

The solution is aligned with SDG 12 and its emphasis on environmentally sound waste management, reuse and recycling. It also has a useful relationship with the life-cycle and continual-improvement principles associated with ISO 14001. The proposed prototype is technically feasible using HTML, CSS and JavaScript and satisfies the core website requirements given in the assessment brief [3].

In future, the system could be expanded into a real e-waste management platform with verified recyclers, location services, QR tracking, databases and institutional dashboards. For the present academic project, the proposed website provides a practical, demonstrable and distinctive IT-based response to improper electronic-waste disposal.

13. References

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